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Approvals :

Job Title	Approved by	Quality Role	Date	Signature status	Comment
Senior Regulatory Affairs Specialist (US)	Jamie Nowacek	Author	12-Jan-2026	Approved	2026 - Jan - 12 6:43a MST
Support Technician	Qimin Hu	Approver	16-Jan-2026	Approved	
GLP Quality Assurance Specialist	Armaan Singh	Approver	16-Jan-2026	Approved	
Director of Lab Operations North America	Erin Lucas	Approver	16-Jan-2026	Approved	

Change Revision History :

Rev.	Chg. #	Date	Change Description
A	1	12-Jan-2026	Initial upload

Document Control / Integrated Management System (IMS) / Diagnexia / Diagnexia Templates/ **Diagnexia Laboratory Validation / Verification for (Record ID) template**

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LABORATORY VALIDATION/VERIFICATION/COMPARABILITY TEMPLATE

PURPOSE: The purpose of this testing is to record and approve test results for outlined test cases. This template should be used with the Diagnexia test plan, DMS-108833.

SCOPE: This test is for Diagnexia USA testing related to Fit for purpose, Intended use, or change control regression / verification testing. It may be adapted for equipment, software, hardware peripherals, or firmware used in **North America** laboratories. It may also include integration testing or customer test method transfer used in a Service offering.

DATE OF TESTING

Description	*Record start and end dates of testing activities.*
START DATE:	
END DATE:	

REFERENCES:

Document ID	Revision used	Document Title
DMS-1111395	A-1	Diagnexia Validation Policy
DMS-108833	A-1	Diagnexia US - Clinical Validation / Verification Plan
DMS-8765	H-1	Diagnexia Procedure for Equipment
DMS-72576	A-1	Diagnexia Master Security Document
DMS-109469	A-1	Diagnexia USA Master Security Document

DMS-8305	E-1	Diagnexia Policy and Procedure for Quality, Governance, and Risk
DMS-8333	G-1	Diagnexia Policy for Secure Software Development Life Cycle (SSDLC)
DMS-77274	B-2	Policy for Automation Development Life Cycle for Automation Products
DMS-69916	B-2	Diagnexia Work Instruction for Integration Configuration including roles and responsibilities
DMS-8338	K-1	Deciphex Procedure for Document Control
DMS-8094	F-1	Diagnexia Policy for Quality Control
DMS-8295	H-1	Deciphex Procedure for Corrective Actions and Improvement
DMS-8335	J-1	Diagnexia Policy and Procedure for Management of Issues, Concerns, Complaints, and Compliments.
DMS-14322	H-2	Diagnexia Procedure for Pathologist Onboarding and Checklist
DMS-41059	D-1	Competency-Based Training for the Quality Control of Digital Images

REGULATORY / QUALITY:

ID	Regulatory / Quality Assurance	Description
1	College of American Pathologist (CAP)	GEN.42195, GEN.43022, GEN.50630, GEN.52920, COM.40475, (COM.40640) - *v2024 Dec 26*
2	CAP Guidelines for WSI validation	https://www.cap.org/protocols-and-guidelines/cap-guidelines/current-cap-guidelines/validating-whole-slide-imaging-for-diagnostic-purposes-in-pathology
3	Clinical Laboratory Amendment Act (CLIA)	42 CFR 493.1253 - Establish Performance Characteristics
4	CLIA Testing Guidance	https://www.cms.gov/regulations-and-guidance/legislation/clia/downloads/6064bk.pdf
5	Good Laboratory Practice	21 CFR Part 58

ID	Regulatory / Quality Assurance	Description
6	Food & Drug Administration (FDA)	21 Part 11 - electronic signature
7	FDA Cure's Act (law)	21st Century Cures Act - determines how to classify medical device(s)
8	FDA GUIDANCE: Medical Device Data Systems - (2022)	Discusses how the FDA does not plan on enforcing devices that only transfer data. The FDA considers "acquiring" data still a device (eg scanner) | Refer to 21 CFR 864.3700
9	FDA GUIDANCE: General Principles of Software Validation - (2002)	Discusses how to validate Consumer Off-the-shelf-software (COTS) and includes change control risk based guidance
10	FDA GUIDANCE: Artificial Intelligence in Software as a Medical Device	Outlines the levels of AI. Because AI tools used in Diagnexia are administrative, the tool is not involved in clinical decisions; clinical patient reporting will still rely on the human. The AI tools used in Diagnexia are sorting and classifying data to make the decision simpler for the human.
11	FDA Statistical Technicals & Sample plans (QIST)	21 CFR 820.250 & https://www.scribbr.com/statistics/normal-distribution/

SYSTEM DESCRIPTION

Automation Tools	*If applicable, list DMS-xxxxx release notes*
Software version(s)	*If applicable, list DMS-xxxxx release notes*
Laboratory	
Laboratory Workstation	*Monitor if applicable*
Operating System	*Windows operating system version, service pack*
Software version - Viewer	
Software version - Portal	
Server location	*(If AWS, list location like Ohio) - https://aws.amazon.com/about-aws/global-infrastructure/regions_az/*
Scanner Manufacturer / Model	
Scanner Serial Number	
Scanner Firmware / Software Version	
Last preventative maintenance	
Integration	

Plug-ins / Extensions / Middleware (if applicable): (Name, version, function, and validation status. If none, state "None.")	
Test Site Location #1	
Test Site Location #2	*Add rows if necessary*
MIRTH version	
File type(s)	
Network configuration description:	
Server location	*https://aws.amazon.com/about-aws/global-infrastructure/regions_az/*
Telepathology	
Test Site location	
Telepathology workstation	
Operating system	
Network Speed test	*If relevant*

INTENDED USE & APPLICATION

Intended Use Description ***(refer to Regulatory Statement)***	*The validation approach is to document the requirements necessary to validate pathologists on the use of the digital pathology review system 'Diagnexia' for primary and secondary consult reports. The approach is designed to be representative of the intended use and should be inclusive of all intended applications (both H&E stained slides from FFPE tissue, IHC and special stains). This includes the scanning of haematoxylin and eosin, immunohistochemistry (IHC) and special stained slides. Hematoxylin and eosin to detect morphology and the presence/absence of tumor cells and immunohistochemistry to detect the presence of specific protein markers to assist with providing an accurate diagnosis after hematoxylin and eosin staining has taken place. This also includes the scanning of special stain slides to detect presence or absence of certain cell types, structures or microorganisms. Each digital image covers the whole slide and images may be viewed, stored, retrieved, and shared allowing the pathologist to make a diagnosis and/or to provide pathological opinions without the need to view the original glass slides through a light microscope.* *The Diagnexia system in this validation study does not utilise image analysis algorithms or automate the interpretation of any stained slide images. Digital images and case data are provided to the pathologist via the Diagnexia system in all situations. The pathologist reviews the case data and generates the pathology report as required. This reflects closely in practice the actual workflow for the pathologist using the Diagnexia system to review digital cases, see diagram below. In this process, glass slides are generated by the Customer/referring institute. The customer scans the slides. Images and case data are accessed by Diagnexia staff, Quality control checked and uploaded to the Diagnexia system. The digital cases are assessed and assigned to the appropriate consultant Pathologist for their assessment. This protocol is not intended to validate WSI & Diagnexia at any external laboratories.* *It is recommended the execution of the validation is completed in the following format. CAP guidance recommends 60 cases to ensure the quality and consistency of the diagnostic performance of digitized slides. 60 cases previously scanned at the Diagnexia laboratory and reviewed by a Diagnexia pathologist are utilized for this study. The original diagnosis for each case is recorded. The cases are then rescanned by the customer utilizing their scanner. The images and case data are accessed by Diagnexia staff, Quality control checked and uploaded to the Diagexia system. The digital cases are assessed and assigned to the appropriate consultant Pathologist for their assessment. The digital diagnosis for each case is recorded.*

Validation type	Commissioning / equipment move Installation Qualification	Operational Qualification Performance Qualification	Validation / new specification or workflow	Verification / regression / change Test Method Other: <describe>
Marketed Country	Canada	U.S.A.	Other Please specify____________________	Global
Modalities	Glass	Digital only	Integration data	Scan glass to digital
Specimen preparation	FFPE, glass	Cytology, glass	Data only	
Stains evaluated	H&E	IHC	Special Stains	
Sample size[2]	<describe the sample size plan>			

TESTERS

Tester / Role #1 - Lead investigator	*Name / Job Title / role in the testing*	Trained: Yes *(list DMS-xxxx)* Not applicable
Tester / Role #2	*Name / Job Title / role in the testing*	Trained: Yes *(list DMS-xxxx)* Not applicable
Tester / Role #3	*Name / Job Title / role in the testing*	Trained: Yes *(list DMS-xxxx)* Not applicable
Tester / Role #4	*Name / Job Title / role in the testing*	Trained: Yes *(list DMS-xxxx)* Not applicable
Tester / Role #5	*Name / Job Title / role in the testing*	Trained: Yes *(list DMS-xxxx)* Not applicable
Tester / Role #6 - independent reviewer	*Name / Job Title / role in the testing*	Trained: Yes *(list DMS-xxxx)* Not applicable

STUDY DESIGN / MEASUREMENT

Measurement[2]	*A concordance of 95% or higher confidence interval is the target for the validation process. It is important to note that the 95% threshold is not a pass/fail benchmark, meaning that if the validation falls below 95%, it does not necessarily mean that the WSI should not be used for patient care. Instead, it indicates that further investigation or improvement is needed to resolve any discrepancies.* *Measurement Clarification:* *Concordance refers to diagnostic agreement between digital and reference diagnoses as assessed by qualified pathologists.* *Testers are responsible for protocol execution and data capture and are not required to have diagnostic skills equivalent to pathologists.*
Acceptance Criteria	• DMS-8094.xx *Diagnexia Policy for Quality Control*
Sensitivity[3] *(Type II)*	• Type II (false negative) - the tester fails to recognize a difference between the <test parameter> when there is a true difference *Sensitivity is the probability that a test will correctly identify true positives (samples with the condition); thus minimizing Type II error (false negatives).*
Specificity[3] *(Type I)*	• Type I (false positive) - the tester identifies a difference between the <test parameter> when really there is no difference *Specificity is the probability that a test will correctly identify true negatives (samples without the condition); thus minimizing Type I error (false positives).*
Accuracy[3]	*Refers to how close the test results are to the actual condition. Measured by repeatability of a single tester, sample, or instrument. [Inter-variability or within variable].*
Precision[3]	*The degree to which a repeated measurement is reproducible between variables; measured by multiple testers, samples, or instruments. [Intra-variability or between variables].*
Deviations	*Unexpected deviations from the test steps are recorded. A disposition of "retest" or "accept as is" with a justification for review is required by the tester so the test may continue. If the test must stop, the tester must alert quality assurance to evaluate next steps.*
Defects	*Unexpected observations or results that are different than what was expected are recorded during test protocol. Refer to:* • *DMS-8335 Diagnexia Policy and Procedure for Management of Issues, Concerns, Complaints, and Compliments.* • *DMS-8017 Procedure for Test Management*

Design of Test:	*Add any workflow diagrams, attachments, or specific test criteria needed to carry out the validation*

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CASE: SAMPLES

ID	Case ID	Organ / Site	Morphology (if relevant)	No. of Slides	Slide Accession ID / Name	Comment
1	R24-3456	Skin	Melanoma, epithelial	1	A	US sample
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						

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ID	Case ID	Organ / Site	Morphology (if relevant)	No. of Slides	Slide Accession ID / Name	Comment
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27						
28						
29						
30						
31						

ID	Case ID	Organ / Site	Morphology (if relevant)	No. of Slides	Slide Accession ID / Name	Comment
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						

ID	Case ID	Organ / Site	Morphology (if relevant)	No. of Slides	Slide Accession ID / Name	Comment
48						
49						
50						
51						
52						
53						
54						
55						
56						
57						
58						
59						
60						

PRE-TEST: Use this section to configure/calibrate equipment, data, or pre-test conditions necessary to conduct the testing.
Pre-Test Requirements:
- All testers must have completed required training and demonstrated competency prior to test execution.
- All equipment must be installed, qualified, and within preventative maintenance per DMS-8765.
- All software must be validated, approved, and operating under change control per DMS-108833.
- User access must be role-based and compliant with security policies.

ID	Test Step / instruction	Expected Result	Evidence Y or N	Actual Result	P/F	Defect ID	Initials / Date (DD-MM-YYYY)

TEST RESULTS

Scoring Criteria:
- Concordant: Digital diagnosis matches reference diagnosis
- Minor Discordance: Difference without clinical impact
- Major Discordance: Difference with potential clinical impact

All discordances must be investigated and documented.

Tester Instructions:
Testers must record actual observed results, deviations, system behavior, and supporting evidence.
Testers must not interpret or assess clinical diagnoses.

Case ID	Expected Result	Actual Result	Tester	Score / concordance	Comment

Case ID	Expected Result	Actual Result	Tester	Score / concordance	Comment
<screen shot evidence>					

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Case ID	Expected Result	Actual Result	Tester	Score / concordance	Comment
<screen shot evidence>					

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Case ID	**Expected Result**	**Actual Result**	**Tester**	**Score / concordance**	**Comment**
<screen shot evidence>					

Case ID	Expected Result	Actual Result	Tester	Score / concordance	Comment

Case ID	Expected Result	Actual Result	Tester	Score / concordance	Comment
<screen shot evidence>					

Document ID: DMS-xxxxxx
Document Title: Diagnexia Laboratory Validation / Verification for (Record ID)
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Case ID	Expected Result	Actual Result	Tester	Score / concordance	Comment
<screen shot evidence>					

CONTINUE TO ADD ROWS

SUMMARIZED FINDINGS

Deviations	<List deviations here> All deviations must be reconciled and dispositioned.
Defects	<Defects must be reported>
Correction / Re-Test	Any re-tests performed, list the step IDs here.
Summation of findings:	
Approval: Medical Laboratory Director	“ *I have reviewed the verification (or validation) data for the performance specifications listed, and the performance of the method is considered acceptable for patient testing.”*

APPROVAL MATRIX

Name/ Role	**e-Signature**	With my signature as an author of the document, I approve the content of the document.
Name/ Role	**e-Signature**	With my signature as an approver of the document, I approve the content of the document.
Name/ Role	**e-Signature**	With my signature as an approver of the document, I approve the content of the document.